Big science and peer review, a game theoretical model

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1 Introduction

2 Model

2.1 Setting

We consider a publication game involving a finite population of N authors and a single academic journal. Our framework aligns with established game-theoretic models of scientific publishing Thurner and Hanel (2011), while specifically building on the incentive structures analyzed by Zollman et al. Zollman et al. (2024). Unlike previous static models, we extend the analysis to account for science volume. The fundamental unit of the game is a research paper, which possesses a binary intrinsic quality $q \in \{0, 1\}$. A paper is defined as "good" ($q = 1$) or "bad" ($q = 0$). In each period, authors make submission decisions, and the journal enforces quality control standards subject to a physical capacity constraint.

The publication process unfolds in a strictly sequential timeline, as illustrated in Figure 1:

1. **Submission (Generation of V):** Based on their strategies, authors decide whether to submit their work. The aggregate collection of all submitted manuscripts forms the *Submission Pool*, denoted by $V \leq N$.

2. **Peer Review (Generation of M):** The journal applies a peer review process to filter the submission pool. Acknowledging the inherent unreliability of assessment documented in empirical studies Bornmann (2011), we model reviewers as imperfect classifiers. They are subject to Type I errors (False Positives, rate λ) and Type II errors (False Negatives, rate ϵ). Papers that successfully pass this screening form the set of *Peer-accepted papers*, denoted by M. It is important to note that membership in M signifies that a paper is "scientifically acceptable," but not necessarily "published."

3. **Capacity Check and Storage (K):** The journal has a fixed resource limit or *Journal Storage*, denoted by K. This constraint reflects the physical limits on publication slots or the administrative burden of processing manuscripts Kovanis et al. (2016).

4. **Final Acceptance (ρ):** The final publication decision depends on the relationship between the volume of peer-accepted papers (M) and the journal storage (K). We define ρ as the *probability of being accepted conditional on being peer-accepted*. This variable captures the congestion effect:

$$\rho = \begin{cases} 1 & \text{if } M \leq K \quad \text{(No Congestion)} \\ K/M & \text{if } M > K \quad \text{(Rationing triggered)} \end{cases} \tag{1}$$

In this framework, a paper is published if and only if it passes two hurdles: the quality hurdle (peer review) and the congestion hurdle (storage constraint).

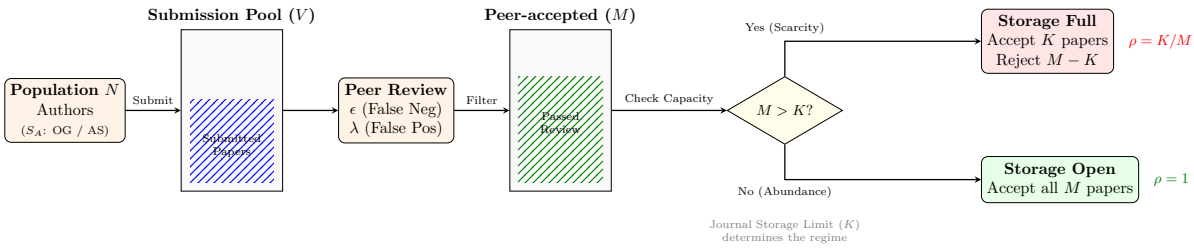


Figure 1: **Schematic representation of the Threshold Competition Model.** The process unfolds in stages: (1) Authors from population N submit papers into pool V; (2) Peer review filters submissions into a set of peer-accepted papers M; (3) The volume M is compared against the journal storage capacity K. If $M > K$, a rationing mechanism is triggered, reducing the conditional acceptance probability ρ below 1.

2.2 Players, strategies, and quality generation

The publication game involves a single journal (J) and a finite population of authors $\mathcal{A} = \{1, 2, \ldots, N\}$. We treat N as a finite exogenous parameter. This finite-population approach allows us to capture the congestion dynamics and "tragedy of the commons" effects inherent in resource-constrained peer review systems, as analyzed in the economic literature on academic publishing (Cotton, 2013; Azar, 2005). Throughout, we use V and M to denote the realised *counts* of submitted and peer-accepted papers, respectively.

Quality generation. In each round, each author generates a manuscript with binary intrinsic quality $q \in \{0, 1\}$, where $q = 1$ denotes a "good" paper and $q = 0$ a "bad" paper. Quality is generated i.i.d. across authors with

$$\Pr(q = 1) = \alpha, \qquad \Pr(q = 0) = 1 - \alpha, \tag{2}$$

where $\alpha \in (0, 1)$ is an exogenous productivity parameter reflecting the baseline scientific capability of the community (Ellison, 2002).

Author strategies (S_A). Each author chooses a submission rule $s_A \in S_A$ that conditions on realised quality. Consistent with signaling models in scientific communication (Tiokhin et al., 2021), we consider four pure strategies:

- **Only Good (OG):** submit iff $q = 1$;
- **Always Submit (AS):** submit regardless of q (strategic spamming);
- **Only Bad (OB):** submit iff $q = 0$ (included for completeness);
- **Never Submit (NS):** never submit.

Equivalently, s_A can be represented by (p_g, p_b), the submission probabilities for good and bad papers:

$$(p_g, p_b) = \begin{cases} (1, 0) & \text{if } s_A = \text{OG} \\ (1, 1) & \text{if } s_A = \text{AS} \\ (0, 1) & \text{if } s_A = \text{OB} \\ (0, 0) & \text{if } s_A = \text{NS}. \end{cases} \tag{3}$$

Journal strategies (S_J). The journal commits to an editorial policy $s_J \in S_J$ that maps the perceived quality signal $\hat{q} \in \{0, 1\}$ (generated by noisy peer review, see Section **??**) into an accept/reject decision:

- **Only Good (OG):** accept iff $\hat{q} = 1$;
- **Always Accept (AA):** accept all submissions (ignores $\hat{q}$);
- **Only Bad (OB):** accept iff $\hat{q} = 0$;
- **Reject All (AR):** reject all submissions (ignores $\hat{q}$).

Exogenous parameters and endogenous variables. To clarify the mechanics of congestion, we distinguish between fixed environmental constraints and emergent outcomes:

- **Exogenous parameters:** Population size (N), productivity (α), journal size (K), review noise parameters (ϵ, λ), and payoff primitives, discussed below: c, cost of submission for authors; r, the reward for an author derived from successful publication; B, the benefit to the journal derived from publishing good papers; and D, the the penalty for journals from publishing bad papers.
- **Endogenous variables:** Submission volume (V), peer-accepted volume (M), the rationing factor ($\rho = \max\left\{\frac{K}{M}, 1\right\}$), and the population state x (strategy shares) governing the equilibrium distribution.

2.3 Review noise and acceptance probabilities

Peer review acts as a stochastic filter between the submission volume V and the peer-accepted volume M. Throughout, *pass* refers to passing the review filter and entering the peer-accepted pool M (not final publication, which is also affected by the journal size.

Noise parameters (ϵ, λ). Consistent with empirical meta-analyses on review reliability (Bornmann et al., 2010; Cicchetti, 1991), we assume reviewers are imperfect classifiers. Let $\hat{q} \in \{0, 1\}$ denote the perceived quality signal for a paper with true quality $q \in \{0, 1\}$. We define:

$$\epsilon = \Pr(\hat{q} = 0 \mid q = 1), \qquad \lambda = \Pr(\hat{q} = 1 \mid q = 0), \tag{4}$$

where ϵ is the false negative rate (acting as a tax on honesty) and λ the false positive rate (acting as an incentive for opportunism), a structure supported by empirical studies on scientific gatekeeping (Siler et al., 2015).

Conditional acceptance probabilities. Given a journal policy s_J, define the probability of acceptance conditional on true quality as:

$$\pi_q(s_J) \equiv \Pr(\text{pass} \mid q, s_J), \qquad q \in \{0, 1\}. \tag{5}$$

Under the journal strategies defined in Section 2.2, and aligned with theoretical models of gatekeeper selection (Muller-Itten, 2022), the acceptance vector (π_1, π_0) is given by:

$$(\pi_1(s_J), \pi_0(s_J)) = \begin{cases} (1-\epsilon,\ \lambda) & \text{if } s_J = \text{OG} \\ (1,\ 1) & \text{if } s_J = \text{AA} \\ (\epsilon,\ 1-\lambda) & \text{if } s_J = \text{OB} \\ (0,\ 0) & \text{if } s_J = \text{AR}. \end{cases} \tag{6}$$

Per-author pass-review probability ($\mathcal{P}$). Let $(p_g(s_A), p_b(s_A))$ denote the submission probabilities conditional on realised quality. We define $\mathcal{P}(s_A, s_J)$ as the *ex-ante* probability that an author playing s_A contributes a paper to the peer-accepted pool M under journal policy s_J:

$$\mathcal{P}(s_A, s_J) = \alpha\, p_g(s_A)\, \pi_1(s_J) + (1-\alpha)\, p_b(s_A)\, \pi_0(s_J). \tag{7}$$

Below we enumerate the effective acceptance probabilities for all author strategies under each journal regime.

Case 1: Standard Review (Journal plays Only Good). This is the baseline scenario where $\pi_1 = 1 - \epsilon$ and $\pi_0 = \lambda$.

$$\mathcal{P}(\text{OG}, \text{OG}) = \alpha(1-\epsilon), \tag{8a}$$
$$\mathcal{P}(\text{AS}, \text{OG}) = \alpha(1-\epsilon) + (1-\alpha)\lambda, \tag{8b}$$
$$\mathcal{P}(\text{OB}, \text{OG}) = (1-\alpha)\lambda, \tag{8c}$$
$$\mathcal{P}(\text{NS}, \text{OG}) = 0. \tag{8d}$$

Case 2: Permissive (Journal plays Always Accept). The screening is ignored ($\pi_1 = 1, \pi_0 = 1$), making acceptance depend solely on submission.

$$\mathcal{P}(\text{OG}, \text{AA}) = \alpha, \tag{9a}$$
$$\mathcal{P}(\text{AS}, \text{AA}) = \alpha + (1 - \alpha) = 1, \tag{9b}$$
$$\mathcal{P}(\text{OB}, \text{AA}) = 1 - \alpha, \tag{9c}$$
$$\mathcal{P}(\text{NS}, \text{AA}) = 0. \tag{9d}$$

Case 3: Inverse Selection (Journal plays Only Bad). The journal accepts papers perceived as bad ($\hat{q} = 0$). Good papers pass only via false negatives ($\pi_1 = \epsilon$), and bad papers pass via true negatives ($\pi_0 = 1 - \lambda$).

$$\mathcal{P}(\text{OG}, \text{OB}) = \alpha\epsilon, \tag{10a}$$
$$\mathcal{P}(\text{AS}, \text{OB}) = \alpha\epsilon + (1 - \alpha)(1 - \lambda), \tag{10b}$$
$$\mathcal{P}(\text{OB}, \text{OB}) = (1 - \alpha)(1 - \lambda), \tag{10c}$$
$$\mathcal{P}(\text{NS}, \text{OB}) = 0. \tag{10d}$$

Case 4: Closed (Journal plays Reject All). No papers enter the peer-accepted pool ($\pi_1 = 0, \pi_0 = 0$).

$$\mathcal{P}(s_A, \text{AR}) = 0 \quad \forall s_A \in S_A. \tag{11}$$

Aggregate peer-accepted volume ($E[M]$). To determine the system-wide congestion, we aggregate these individual passing probabilities across the entire population, a method consistent with burden estimation in peer review (Kovanis et al., 2016). Let $\mathbf{x} = \{x_{s_A} \mid s_A \in S_A\}$ denote the **population state**, where $x_{s_A} \in [0, 1]$ represents the proportion of authors adopting strategy s_A, satisfying $\sum_{s_A \in S_A} x_{s_A} = 1$.

The expected total volume of peer-accepted papers, $E[M]$, is the sum of contributions from all strategy groups:

$$E[M] = \sum_{i=1}^{N} \mathcal{P}(s_{A,i}, s_J) = N \sum_{s_A \in S_A} x_{s_A} \mathcal{P}(s_A, s_J). \tag{12}$$

This aggregate volume $E[M]$ is the critical endogenous variable that interacts with the journal capacity K to determine the rationing probability ρ defined in Eq. (1).

2.4 Capacity constraint and rationing probability (ρ)

A key feature of the model is the separation between *scientific merit* (passing peer review and entering the set of peer-accepted papers) and *publication slots* (capacity-constrained publication).

Journal capacity (K). Let $K \in \mathbb{N}^+$ denote the journal's exogenous publication capacity: at most K papers can be published per cycle due to limited pages and editorial resources.

Uniform rationing and the rationing probability (ρ). Let M be the realised number of peer-accepted papers in a given round. If $M \leq K$, all peer-accepted papers are published. If $M > K$, the journal applies *uniform rationing*: it selects exactly K papers uniformly at random from the M peer-accepted papers. Hence, the conditional publication probability for a peer-accepted paper is determined by the realised congestion:

$$\rho(M) \equiv \Pr(\text{published} \mid \text{pass}) = \min\left(1, \frac{K}{M}\right) = \begin{cases} 1 & \text{if } M \leq K, \\ \frac{K}{M} & \text{if } M > K. \end{cases} \tag{13}$$

Ex-post vs. ex-ante congestion. Across rounds, the peer-accepted count M is stochastic because it aggregates individual submission and review outcomes. Authors therefore form expectations over congestion given the population state $\mathbf{x}$ and journal policy s_J, implying an expected rationing factor:

$$\mathbb{E}[\rho] = \mathbb{E}\left[\min\left(1, \frac{K}{M}\right)\right]. \tag{14}$$

Crucially, increases in low-quality or opportunistic submissions raise the expected volume $\mathbb{E}[M]$ and mechanically reduce $\mathbb{E}[\rho]$, generating a negative congestion externality on all authors.

The journal chooses one of four editorial strategies:

$$S_J = \{OA, OG, OB, NS\},$$

defined as acceptance rules based on perceived paper quality:

- OA (*Open/Always Accept*): accept all submissions;
- OG (*Only Good*): accept submissions perceived as good;
- OB (*Only Bad*): accept submissions perceived as bad;
- NS (*No Select/Reject All*): reject all submissions.

2.5 Payoff analysis

In this subsection, we derive the expected payoffs for both authors and the journal. The introduction of the capacity constraint K couples the payoffs: an individual's utility depends not only on their own strategy but also on the aggregate congestion generated by the population.

2.5.1 Notation and general framework

Table 1: Summary of key parameters and variables used in the payoff analysis.

Symbol	**Description**
Exogenous Parameters (Fixed)	
N	Total number of authors
K	Journal capacity (maximum publication slots per cycle)
α	Probability that an author produces a good paper
ϵ	False negative rate (Prob. of rejecting a good paper)
λ	False positive rate (Prob. of accepting a bad paper)
r	Reward for a published paper (Author)
c	Cost of submission (Author)
B	Benefit for publishing a good paper (Journal)
D	Penalty for publishing a bad paper (Journal)
C_J	Unit cost of peer review (per submission)
Endogenous Variables (Dependent on Strategy)	
V	Total number of submissions (Submission Pool)
M	Number of papers passing peer review (Perception Pool)
Y	Number of papers finally published ($Y \leq K$)
$\bar{\rho}$	Expected rationing probability ($\mathbb{E}[\min(1, K/M)]$)
x	Proportion of authors adopting a specific strategy (Population State)

We first establish a unified notation. To ensure analytical tractability while capturing the congestion effect, we employ a mean-field approximation where players base their decisions on the **expected** state of the system.

Submission and publication probabilities. For an author playing strategy $s_A \in S_A$, we define the unconditional probability of submitting a paper as $P_{\text{sub}}(s_A)$. Based on the submission vectors (p_g, p_b) defined in Section 2.2, this is:

$$P_{\text{sub}}(s_A) = \alpha p_g(s_A) + (1-\alpha) p_b(s_A). \tag{15}$$

Recall from Section 2.3 that $\mathcal{P}(s_A, s_J)$ denotes the probability of *entering the peer-accepted pool* M. However, entering M does not guarantee publication. Let $\bar{\rho}$ denote the **expected rationing probability**:

$$\bar{\rho} \equiv \mathbb{E}[\rho] = \mathbb{E}\left[\min\left(1, \frac{K}{M}\right)\right]. \tag{16}$$

The joint probability of a paper being finally published is the product of passing review and passing the capacity check: $\bar{\rho} \cdot \mathcal{P}(s_A, s_J)$.

Volumes: Submitted (V), Passed (M), and Published (Y). It is crucial to distinguish between papers that pass peer review (M) and those that are finally published (Y). Let V_G and V_B be the expected volumes of good and bad submissions generated by the population. Let M_G and M_B be the expected volumes of papers entering the peer-accepted pool. We define Y_G and Y_B as the expected volumes of **finally published** papers:

$$Y_G = \bar{\rho} M_G, \quad Y_B = \bar{\rho} M_B. \tag{17}$$

2.5.2 General payoff functions

Author's utility (U_A). Let r be the reward for a published paper and c be the cost of submission. The expected payoff for an author playing s_A against a journal s_J is the expected reward from publication minus the reliable cost of submission:

$$U_A(s_A, s_J) = r \cdot \underbrace{\bar{\rho} \cdot \mathcal{P}(s_A, s_J)}_{\text{Prob(Published)}} - c \cdot \underbrace{P_{\text{sub}}(s_A)}_{\text{Prob(Submitted)}} \tag{18}$$

Journal's utility (U_J). Let B be the benefit of publishing a good paper, D be the penalty for a bad paper, and C_J be the unit cost of reviewing a submission. The journal's utility depends on the *published* volumes (Y_G, Y_B) and the *submitted* volume $(V = V_G + V_B)$:

$$U_J(s_J) = B \cdot Y_G - D \cdot Y_B - C_J \cdot V \tag{19}$$

Substituting $Y = \bar{\rho} M$, we obtain the working form of the journal's payoff:

$$U_J(s_J) = \bar{\rho}\left[B \cdot M_G(s_J) - D \cdot M_B(s_J)\right] - C_J \cdot V \tag{20}$$

Remark: Eq. (20) assumes $\mathbb{E}[\rho M] \approx \mathbb{E}[\rho]\mathbb{E}[M]$. This mean-field approximation is standard in evolutionary game theory for large finite populations.

2.5.3 Derivation of strategy-specific payoffs

We now explicitly list the payoff functions for the 8 primary scenarios (4 Author strategies and 4 Journal strategies).

Author payoffs (under Journal Strategy s_J = OG). In the standard academic context, the journal plays **Only Good (OG)**. The review filter applies $(\pi_1, \pi_0) = (1-\epsilon, \lambda)$. Substituting the specific $\mathcal{P}$ and P_{sub} values into Eq. (18), we derive:

$$U_A(\text{OG}, \text{OG}) = \alpha\left[r\bar{\rho}(1-\epsilon) - c\right] \tag{21}$$

$$U_A(\text{AS}, \text{OG}) = r\bar{\rho}\left[\alpha(1-\epsilon) + (1-\alpha)\lambda\right] - c \tag{22}$$

$$U_A(\text{OB}, \text{OG}) = (1-\alpha)\left[r\bar{\rho}\lambda - c\right] \tag{23}$$

$$U_A(\text{NS}, \text{OG}) = 0 \tag{24}$$

Journal payoffs (under Population State x). The journal chooses a strategy s_J to maximize utility given the incoming flow of submissions. Let $V_G(\mathbf{x})$ and $V_B(\mathbf{x})$ be the volumes of good and bad submissions determined by the population state.

1. Only Good (OG): The journal filters papers. The passed volumes are $M_G = V_G(1-\epsilon)$ and $M_B = V_B\lambda$.

$$U_J(\text{OG}) = \bar{\rho}\,[BV_G(1-\epsilon) - DV_B\lambda] - C_J(V_G + V_B) \tag{25}$$

2. Always Accept (AA): The journal accepts everything ($M_G = V_G, M_B = V_B$). Congestion is maximized here, so $\bar{\rho}$ will be minimal.

$$U_J(\text{AA}) = \bar{\rho}\,[BV_G - DV_B] - C_J(V_G + V_B) \tag{26}$$

3. Only Bad (OB): The journal applies inverse selection ($M_G = V_G\epsilon, M_B = V_B(1-\lambda)$).

$$U_J(\text{OB}) = \bar{\rho}\,[BV_G\epsilon - DV_B(1-\lambda)] - C_J(V_G + V_B) \tag{27}$$

4. Reject All (AR): The journal publishes nothing ($Y = 0$). Review costs are still incurred for processed submissions.

$$U_J(\text{AR}) = -C_J(V_G + V_B) \quad \text{(normalized to 0 in dominance analysis)} \tag{28}$$

2.5.4 Summary of transition probabilities

For completeness, Table 2 summarizes the effective probability $\mathcal{P}(s_A, s_J)$ of entering the peer-accepted pool for any strategy pair. These values are the key inputs for the payoff equations.

Table 2: Matrix of Effective Passing Probabilities $\mathcal{P}(s_A, s_J)$. This represents the probability that an author's paper enters the peer-accepted pool M (before rationing).

Author \ Journal	**Only Good (OG)**	**Always Accept (AA)**	**Only Bad (OB)**	**Reject All (AR)**
OG (α good)	$\alpha(1-\epsilon)$	α	$\alpha\epsilon$	0
AS (All)	$\alpha(1-\epsilon) + (1-\alpha)\lambda$	1	$\alpha\epsilon + (1-\alpha)(1-\lambda)$	0
OB ($1-\alpha$ bad)	$(1-\alpha)\lambda$	$1-\alpha$	$(1-\alpha)(1-\lambda)$	0
NS (None)	0	0	0	0

2.6 Dominance analysis and reduced game

Before analyzing the evolutionary dynamics, we simplify the strategy space by iteratively eliminating strictly dominated strategies. We conduct this analysis under the standard **mean-field approximation**, holding the aggregate congestion factor $\bar{\rho}$ fixed. The elimination proceeds in two stages: first, we identify strategies that are dominated purely by the structural logic of the game (independent of specific cost-benefit inequalities); second, we introduce parameter constraints to eliminate strategies based on economic trade-offs.

2.6.1 Stage 1: Structural elimination (Unconditional dominance)

We first establish dominance relations that hold under weak axiomatic assumptions regarding the nature of peer review and scientific value.

Elimination of Journal "Only Bad" (OB). The strategy **Only Bad (OB)** represents an inverse selection policy. We compare it to **Only Good (OG)**. Using the payoff functions derived in Section 2.5, the difference in expected utility is:

$$\begin{aligned}\Delta U_J &= U_J(\text{OG}) - U_J(\text{OB}) \\ &= \bar{\rho}\,[BV_G(1-\epsilon) - DV_B\lambda] - \bar{\rho}\,[BV_G\epsilon - DV_B(1-\lambda)] \\ &= \bar{\rho}\,[BV_G(1-2\epsilon) + DV_B(1-2\lambda)]\end{aligned} \tag{29}$$

Condition: We assume peer review is **informative**, defined as performing better than random guessing:

$$\epsilon < 0.5 \quad \text{and} \quad \lambda < 0.5. \tag{30}$$

Under this definition, both $(1-2\epsilon)$ and $(1-2\lambda)$ are strictly positive. Since $B, D, V_G, \bar{\rho} > 0$, it follows that $\Delta U_J > 0$. Thus, **Journal OB is strictly dominated by OG** solely by virtue of the definition of peer review quality, regardless of the magnitude of rewards or penalties.

Elimination of Author "Only Bad" (OB). Consider the choice between **Always Submit (AS)** and **Only Bad (OB)**. AS submits both paper types ($q \in \{0,1\}$), while OB submits only bad papers ($q = 0$). The payoff difference is the marginal utility of submitting a good paper:

$$\begin{aligned}\Delta U_A &= U_A(\text{AS}, \text{OG}) - U_A(\text{OB}, \text{OG}) \\ &= [U_A(q=1) + U_A(q=0)] - U_A(q=0) \\ &= U_A(q=1) = \alpha\,[r\bar{\rho}(1-\epsilon) - c]\end{aligned} \tag{31}$$

Condition: We assume the **Axiom of Value**: in a functional scientific system, a correctly identified high-quality paper yields a positive net return (i.e., $r(1-\epsilon) > c$). If this axiom holds, then $\Delta U_A > 0$ for any non-zero congestion $\bar{\rho}$. Thus, **Author OB is strictly dominated by AS** due to monotonicity: a rational author will never withhold a high-quality paper while simultaneously submitting a low-quality one.

2.6.2 Parameter constraints for a non-trivial game

Having eliminated the structurally irrational strategies, we now focus on the remaining set. To ensure the game yields interesting evolutionary dynamics (avoiding trivial collapse or utopian perfection), we define the "viable region" of the parameter space using the **abundance benchmark** ($\bar{\rho} = 1$).

1. **Viability Constraint (Honesty is profitable):**

$$r(1-\epsilon) > c \tag{32}$$

   This formally restates the Axiom of Value. It ensures that without congestion, the honest strategy (OG) dominates silence (NS).

2. **Profitability Constraint (Spamming is profitable):**

$$\lambda r > c \tag{33}$$

   This constraint drives the congestion dilemma. It implies that in a frictionless system ($\bar{\rho} = 1$), the expected reward of a bad paper exceeds the submission cost. If $c > \lambda r$, bad papers are never submitted, and the game trivially converges to perfect honesty.

3. **Reputation Constraint (Screening is preferred):**

$$DV_B(1-\lambda) > BV_G\epsilon \tag{34}$$

   This condition ensures the journal prefers screening (OG) over blind acceptance (AA). It states that the aggregate penalty avoided by rejecting bad papers outweighs the benefit lost from accidentally rejecting good ones.

2.6.3 Stage 2: Conditional elimination (Economic dominance)

We now eliminate strategies based on the specific economic trade-offs defined above.

Dominance of Journal "Always Accept" (AA). We compare **Only Good (OG)** with **Always Accept (AA)**. Assuming the processing cost C_J applies to all submissions (Processing Cost Model), the difference depends on the net value of papers rejected by OG but accepted by AA:

$$\begin{aligned}U_J(\text{OG}) - U_J(\text{AA}) &= \bar{\rho}\left[(BM_G^{\text{OG}} - DM_B^{\text{OG}}) - (BM_G^{\text{AA}} - DM_B^{\text{AA}})\right] \\ &= \bar{\rho}\left[D(V_B - M_B^{\text{OG}}) - B(V_G - M_G^{\text{OG}})\right] \\ &= \bar{\rho}\,[DV_B(1-\lambda) - BV_G\epsilon]\end{aligned} \tag{35}$$

Under the **Reputation Constraint** (34), the penalty avoided (first term) exceeds the benefit lost (second term). Thus, **AA is strictly dominated by OG**.

Dominance of Journal "Reject All" (AR). Strategy AR yields a normalized payoff of 0 (or $-C_J V$). Since OG generates positive utility from good papers ($B > 0$) and filters bad ones, OG dominates AR provided the net value of the peer-accepted pool is positive: $BM_G > DM_B$. We assume the journal operates in a parameter region where valid science is net positive.

Dominance of Author "Never Submit" (NS). Comparing **Only Good (OG)** with **Never Submit (NS)** ($U_A = 0$):

$$U_A(\text{OG}, \text{OG}) - U_A(\text{NS}, \text{OG}) = \alpha \left[r\bar{\rho}(1-\epsilon) - c \right] \tag{36}$$

Under the **Viability Constraint** (32), this difference is positive as long as the congestion does not exceed the critical threshold $\bar{\rho} > \frac{c}{r(1-\epsilon)}$. In the viable regime, **NS is strictly dominated by OG** (and by AS).

2.6.4 The reduced evolutionary game

Following the two-stage elimination process, the complex multi-strategy interaction reduces to a concise 2×1 asymmetric game.

1. **The Journal** is a fixed institutional player locked into the **Only Good (OG)** strategy. It acts as the environment, applying a filter (ϵ, λ) and a capacity constraint K.
2. **The Author Population** evolves between two surviving pure strategies:
   - **Only Good (OG) [Cooperators]:** Authors who submit only high-quality work ($q = 1$).
   - **Always Submit (AS) [Defectors]:** Authors who submit everything ($q \in \{0, 1\}$), effectively spamming the journal with bad papers to exploit λ.

We define the **population state** $x(t) \in [0, 1]$ as the fraction of authors adopting the **Only Good** strategy. The evolutionary dynamics of the system are thus entirely determined by the time evolution of x.

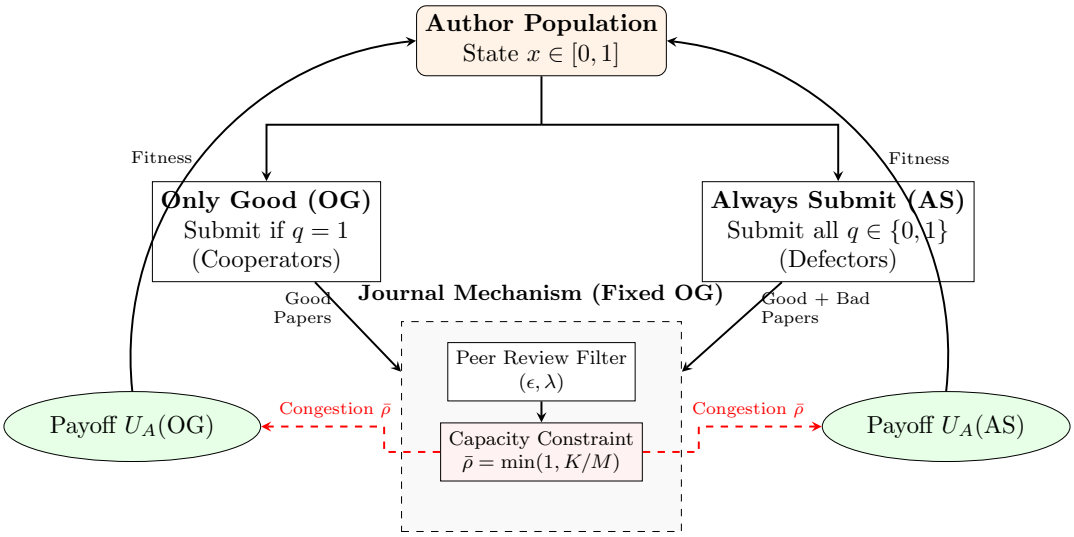


Figure 2: **Structure of the Reduced Evolutionary Game.** The author population evolves between two strategies: Only Good (OG) and Always Submit (AS). The Journal acts as a fixed mechanism consisting of a peer review filter and a capacity constraint. Submissions from AS authors increase the total volume M, triggering the capacity constraint. The resulting **Congestion Externality** ($\bar{\rho}$) reduces the publication probability for all papers, feeding back into the payoffs and driving the evolutionary dynamics.

3 Replicator Dynamics and Bifurcation Analysis

3.1 Model reduction and population state

Following the dominance analysis, the strategy space reduces to a dynamic contest between two surviving Author strategies, with the Journal playing a fixed filtering role. We define the **population state** $x(t) \in [0, 1]$ as the proportion of authors playing **Only Good (OG)**, while $1 - x(t)$ play **Always Submit (AS)**.

The evolutionary trajectory is governed by the payoff differential. We assume the system operates in the **Conditional Profitability Region**:

$$\lambda r < c < r \tag{37}$$

This condition implies that in the un-congested limit ($\eta \approx 1, \bar{\rho} \approx 1$), cheating is unprofitable ($\lambda r - c < 0$). Cheating becomes viable only if the screening mechanism collapses significantly due to congestion.

3.2 Endogenous congestion functions

To capture the dual impact of congestion, we model two distinct bottlenecks. While numerical simulations use a smoothed approximation (Soft Minimum) to capture pre-congestion friction, we present the analytical forms using standard cutoffs for clarity.

3.2.1 Stage 1: The screening intensity function $\eta(x)$

The total submission volume is $V(x) = N[1 - (1 - \alpha)x]$. The screening intensity $\eta(x)$ represents the probability that a paper undergoes rigorous peer review. It degrades when volume exceeds the review capacity K_{rev}:

$$\eta(x) = \min\left(1, \frac{K_{rev}}{V(x)}\right) \tag{38}$$

We assume "rubber-stamping" implies unconditional acceptance of unscreened papers. Thus, the pass probability for a bad paper, $\pi_0(x)$, increases as screening fails:

$$\pi_0(x) = \eta(x)\lambda + (1 - \eta(x)) \cdot 1 = 1 - \eta(x)(1 - \lambda) \tag{39}$$

For a good paper, the pass probability $\pi_1(x)$ also approaches 1 as scrutiny relaxes (assuming false negatives ϵ only occur during rigorous review):

$$\pi_1(x) = \eta(x)(1 - \epsilon) + (1 - \eta(x)) \cdot 1 = 1 - \eta(x)\epsilon \tag{40}$$

3.2.2 Stage 2: The rationing probability function $\bar{\rho}(x)$

Let K be the publication slot capacity. The expected volume of papers entering the "perceived good" pool $M(x)$ is:

$$\begin{aligned} M(x) &= V_G(x)\pi_1(x) + V_B(x)\pi_0(x) \\ &= N\alpha \cdot \pi_1(x) + N(1 - \alpha)(1 - x) \cdot \pi_0(x) \end{aligned} \tag{41}$$

Using a mean-field approximation, the expected rationing probability is:

$$\bar{\rho}(x) \approx \min\left(1, \frac{K}{M(x)}\right) \tag{42}$$

3.3 Derivation of the replicator equation

Since both OG and AS strategies submit good papers whenever they occur (with identical acceptance probabilities), the expected payoff component from good papers cancels out in the differential. The dynamics are driven entirely by the net utility of the additional "bad" papers submitted by AS authors, denoted as $\Pi_{bad}(x)$. The replicator equation is:

$$\dot{x} = -x(1 - x)(1 - \alpha)\Pi_{bad}(x) \tag{43}$$

A bad paper yields reward r only if it passes both bottlenecks. The expected net utility is:

$$\Pi_{bad}(x) = r \cdot \bar{\rho}(x) \cdot \pi_0(x) - c \tag{44}$$

3.4 Bifurcation and stability analysis

The system exhibits **Bistability** because the two congestion effects work in opposition:

- **Review Collapse ($\pi_0 \uparrow$):** Increases the incentive to cheat.
- **Storage Rationing ($\bar{\rho} \downarrow$):** Decreases the incentive to cheat.

When the review collapse effect dominates, $\Pi_{bad}(x)$ becomes positive at low x, creating a corruption trap.

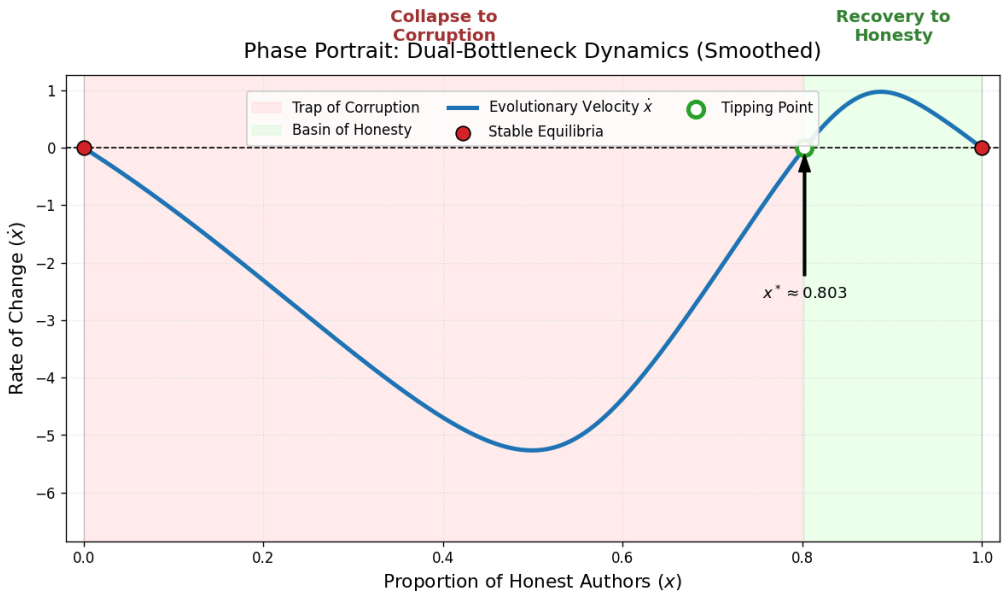


Figure 3: **Phase Portrait of the Dual-Bottleneck Dynamics.** The curve shows the evolutionary velocity $\dot{x}$ using a smoothed simulation model. The system exhibits three equilibria: (1) **Trap of Corruption** ($x = 0$): Stable, as review collapse makes spamming profitable. (2) **Basin of Honesty** ($x = 1$): Stable, as strict screening deters spamming. (3) **Tipping Point** ($x^* \approx 0.803$): An unstable internal root. Initial states $x > x^*$ recover to honesty, while $x < x^*$ collapse.

3.4.1 Analytical Condition for Instability of the Internal Equilibrium

To rigorously characterize the stability of the internal equilibrium x^*, we analyze the derivative of the velocity field. Since $\dot{x} \propto -\Pi_{bad}(x)$, the equilibrium x^* is **unstable** (a repeller) if and only if:

$$\left.\frac{d\Pi_{bad}}{dx}\right|_{x^*} < 0 \tag{45}$$

Using the chain rule via submission volume V:

$$\frac{d\Pi_{bad}}{dx} = \frac{d\Pi_{bad}}{dV} \cdot \frac{dV}{dx} \tag{46}$$

We know $\frac{dV}{dx} = -N(1-\alpha) < 0$. Thus, instability requires $\frac{d\Pi_{bad}}{dV} > 0$. Expanding $\Pi_{bad}(V) = r\bar{\rho}(V)\pi_0(V) - c$:

$$\frac{d\Pi_{bad}}{dV} = r\left(\bar{\rho}(V)\frac{d\pi_0}{dV} + \pi_0(V)\frac{d\bar{\rho}}{dV}\right) \tag{47}$$

Here, $\frac{d\pi_0}{dV} > 0$ (Review Collapse helps cheating) and $\frac{d\bar{\rho}}{dV} < 0$ (Rationing hurts cheating). Therefore, the tipping point is unstable if and only if the **Review-Dominated Condition** holds:

$$\underbrace{\bar{\rho}(V)\frac{d\pi_0}{dV}}_{\text{Gain from lax review}} > \underbrace{\left|\pi_0(V)\frac{d\bar{\rho}}{dV}\right|}_{\text{Loss from rationing}} \tag{48}$$

This mathematically confirms that bistability arises when the degradation of peer review quality outpaces the tightening of publication slots. If this condition holds, $\Pi_{bad}(x)$ is monotonically decreasing in x, forcing the dynamics to bifurcate.

3.5 The regime of stable coexistence: Rationing-dominated dynamics

While the default scenario exhibits bistability (a "tipping point" dynamic), a fundamentally different topological regime emerges when the storage constraint (K) is significantly tighter than the review constraint (K_{rev}). Figure 4 illustrates this scenario, where the evolutionary velocity $\dot{x}$ transitions from positive to negative, creating a **Stable Internal Equilibrium** ($x^* \approx 0.49$).

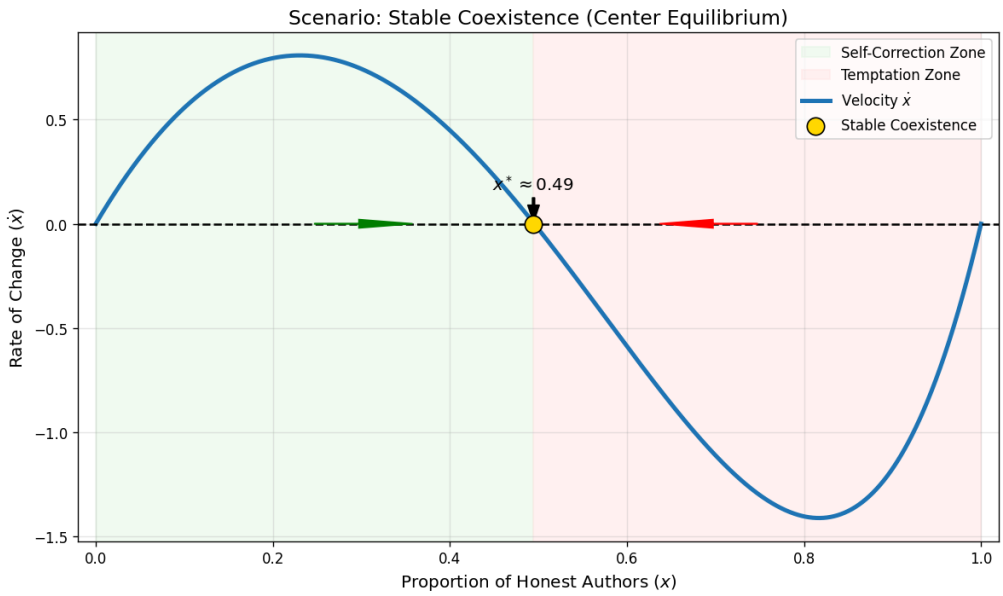


Figure 4: **Phase Portrait of Stable Coexistence (Rationing-Dominated).** The system exhibits a stable attractor at $x^* \approx 0.49$. **Left Zone (Self-Correction):** At low honesty ($x < x^*$), the massive volume of spam triggers severe storage rationing ($\bar{\rho} \to 0$), reducing the expected payoff of cheating below zero. This forces the population to become more honest. **Right Zone (Temptation):** At high honesty ($x > x^*$), the low volume relaxes rationing ($\bar{\rho} \to 1$), making speculative cheating profitable again due to the base false-positive rate λ. The system settles into a permanent mix of honest and opportunistic authors.

3.5.1 Mathematical derivation of the stability condition

For an internal equilibrium x^* to be **stable** (an attractor), the dynamics must be restorative: positive when $x < x^*$ and negative when $x > x^*$. Mathematically, this requires the slope of the velocity field to be negative at the equilibrium:

$$\left.\frac{d\dot{x}}{dx}\right|_{x^*} < 0 \tag{49}$$

Since $\dot{x} \propto -\Pi_{bad}(x)$, this implies that the net utility of cheating $\Pi_{bad}(x)$ must be an **increasing function** of honesty x (or equivalently, a decreasing function of submission volume V). Using the chain rule, we derive the necessary condition:

$$\frac{d\Pi_{bad}}{dx} = \underbrace{\frac{d\Pi_{bad}}{dV}}_{\text{Net Congestion Effect}} \cdot \underbrace{\frac{dV}{dx}}_{<0} > 0 \tag{50}$$

Since $\frac{dV}{dx}$ is strictly negative, the condition for stability requires $\frac{d\Pi_{bad}}{dV} < 0$. Expanding the utility gradient with respect to volume:

$$\frac{d\Pi_{bad}}{dV} = r \left(\underbrace{\frac{d\bar{\rho}}{dV}\pi_0}_{\text{Marginal Rationing Loss}} + \underbrace{\bar{\rho}\frac{d\pi_0}{dV}}_{\text{Marginal Screening Gain}} \right) < 0 \tag{51}$$

Rearranging terms, we obtain the **Rationing-Dominated Condition**:

$$\underbrace{\left| \pi_0(V)\frac{d\bar{\rho}}{dV} \right|}_{\text{Penalty from Crowding}} > \underbrace{\bar{\rho}(V)\frac{d\pi_0}{dV}}_{\text{Benefit from Chaos}} \tag{52}$$

3.5.2 Physical interpretation

This inequality describes a system where the "punishment" from overcrowding (the drop in $\bar{\rho}$) outweighs the "opportunity" from collapsed screening (the rise in π_0).

1. **The Self-Correction Mechanism ($x \to 0$):** When the system is flooded with spam ($x \approx 0$), the submission volume V is maximized. In this regime, if the publication capacity K is sufficiently small (tight storage bottleneck), the rationing probability $\bar{\rho} \approx K/V$ becomes infinitesimal. Even if the peer review system completely fails ($\pi_0 \approx 1$), the probability of securing a publication slot is so low that the expected reward $r \cdot \bar{\rho}$ fails to cover the submission cost c. Thus, $\Pi_{bad} < 0$, and rational actors switch to the honest strategy ($\dot{x} > 0$).

2. **The Temptation Mechanism ($x \to 1$):** When the system is clean ($x \approx 1$), volume is low. Rationing is relaxed ($\bar{\rho} \approx 1$), and screening is functional but imperfect ($\pi_0 = \lambda$). If the parameters satisfy the conditional profitability condition ($\lambda r > c$), cheating becomes profitable. Thus, $\Pi_{bad} > 0$, and actors switch to the opportunistic strategy ($\dot{x} < 0$).

The system is therefore trapped between the "Scylla" of unprofitable congestion and the "Charybdis" of profitable temptation, settling into a stable equilibrium of coexistence where the marginal entrant is indifferent between honesty and spamming.

3.5.3 Real-world implication: The "Defense by Congestion" paradox

The emergence of a stable internal equilibrium ($x^* \approx 0.49$) offers profound insights into the resilience and vulnerability of academic ecosystems. Unlike the bistable regime where the system collapses completely, this regime represents a state of **permanent mediocrity**—a "tolerable" level of pollution that the system sustains indefinitely.

1. **The "Defense by Congestion" Mechanism:** The self-correction observed at low honesty ($x < x^*$) reveals a paradoxical phenomenon: **extreme congestion acts as an unintentional defense mechanism**. In prestigious journals with strict page limits ($K \ll V$), even if peer review fails completely (rubber-stamping), the sheer volume of submissions drives the acceptance rate ($\bar{\rho}$) so low that "spamming" becomes statistically unprofitable. This explains why top-tier journals do not degenerate into full corruption even when their review processes are strained—the "lottery" is simply too hard to win, forcing authors to self-regulate.

2. **The Vulnerability of "Open" Systems:** Conversely, the temptation observed at high honesty ($x > x^*$) warns against complacency in well-functioning systems. When a journal is efficient and uncrowded ($\bar{\rho} \approx 1$), the only barrier to entry is peer review quality. If this quality is merely

"good but not perfect" ($\lambda > c/r$), the system is inherently unstable. This suggests that **capacity expansion** (increasing K)—often seen in the shift to Mega-Journals—can destabilize a healthy ecosystem. By removing the storage bottleneck, the system loses its passive defense against spam, potentially shifting the topology from stable coexistence to a catastrophic collapse (the bistable regime).

In summary, the stable coexistence curve suggests that physical constraints (page budgets), while often criticized for slowing down science, play a hidden structural role in stabilizing the incentive landscape against total corruption.

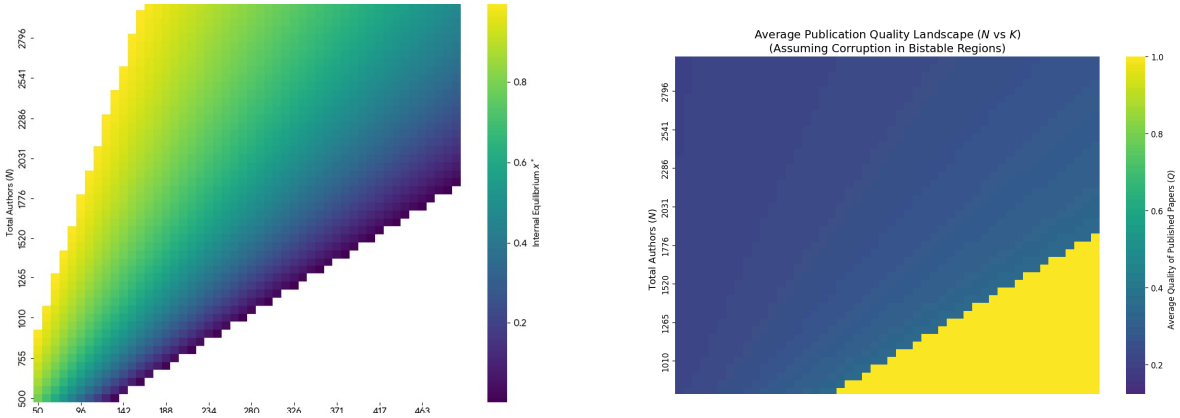
4 Joint Impact of Population and Capacity on Publication Quality

In this section, we extend our analysis from the internal stability equilibrium (x^*) to the macroscopic observable of the academic ecosystem: the *Average Quality of Published Papers* (Q). By varying the total author population (N) and journal capacity (K) simultaneously, we construct a phase diagram that reveals the structural limits of the peer-review system. To map this quality landscape, we first determine the final evolutionary state of the population, denoted as x_{final}, based on the previously derived tipping point analysis. The system's behavior can be categorized into three distinct regimes based on the stability of the equilibrium. In the Global Honesty Regime ($x^* < 0$), the system possesses a single stable attractor at $x = 1$, implying that the final state is one of full honesty ($x_{final} = 1$). Conversely, in the Global Corruption Regime ($x^* > 1$), the single stable attractor lies at $x = 0$, leading to a final state of full corruption ($x_{final} = 0$). The intermediate Bistability Regime ($0 \leq x^* \leq 1$) presents a more complex scenario with two stable attractors. To evaluate the structural risk posed by congestion in this region, we adopt a "Worst-Case Scenario" approach, assuming that stochastic perturbations or insufficient initial trust will inevitably drive the system toward the corrupt attractor. Consequently, for the purpose of risk mapping, we assign $x_{final} = 0$ throughout the bistable region. Based on the determined final state x_{final}, we calculate the Average Quality (Q) of the accepted papers, defined as the ratio of valid scientific contributions to the total volume of accepted literature:

$$Q(x_{final}) = \frac{V_{good}^{acc}}{V_{good}^{acc} + V_{bad}^{acc}} \tag{53}$$

The volume of accepted papers depends critically on the submission behavior and the screening intensity σ. We assume that honest authors produce good papers at a rate α. Since the review process targets logical fallacies and misconduct rather than legitimate research, good papers are assumed to pass screening with probability 1, yielding a volume of $V_{good}^{acc} = N\alpha$. In contrast, opportunistic authors submit bad papers at a rate $(1-\alpha)$. These papers are accepted only if they successfully evade the screening process. The probability of evasion is given by $(1-\sigma)$, where $\sigma = \min\left(1, \frac{K}{C_J V_{total} + \epsilon}\right)$ is the realized screening probability constrained by capacity K and the total submission volume. Thus, the volume of accepted bad papers is expressed as $V_{bad}^{acc} = N(1-\alpha)(1-x_{final})(1-\sigma)$. We present the comparative analysis of the stability landscape and the resulting quality landscape in Figure 5. Figure 5(a) displays the internal equilibrium x^*, identifying the boundaries of stability, while Figure 5(b) visualizes the resulting average publication quality. A striking feature of the quality landscape is the sharp, non-smooth discontinuity where the average publication quality plunges abruptly from a perfect score of 1.0 to a low plateau of approximately 0.4. This "cliff-like" behavior represents a Phase Transition of the academic ecosystem, occurring when a macroscopic property of the system changes discontinuously as a control parameter crosses a critical threshold.

The mechanism driving this phase transition can be understood by examining the system's response to congestion. In the "Safe Zone," where capacity K is sufficient relative to population N, the system's attractor is $x_{final} = 1$. The submission pool consists exclusively of good papers, maintaining a perfect quality score of $Q = 1.0$. However, the moment the population N exceeds the critical carrying capacity defined by the contour $x^* = 0$, the system undergoes a catastrophic collapse, flipping to the corrupt attractor $x_{final} = 0$. This microscopic jump in strategy triggers a macroscopic explosion in submission volume from $N\alpha$ to N. In this corrupt state, the massive volume overwhelms the journal capacity, significantly diluting the screening probability σ. Consequently, the quality does not decay gradually to zero but settles at a "Diluted Limit" determined by the ratio of signal to noise. For instance, with $\alpha = 0.2$



(a) Stability Landscape (x^*)

(b) Quality Landscape (Q)

Figure 5: Joint analysis of system stability and publication quality under varying Population (N) and Capacity (K). (a) The stability landscape shows the transition from Global Honesty (Green/Blue) to Global Corruption (Red) via a Bistable region. (b) The quality landscape reveals a sharp **Phase Transition**. The quality remains at a perfect 1.0 (Green) until the critical threshold is crossed, causing an instantaneous drop to a low plateau of ≈ 0.4 (Red), signifying the catastrophic collapse of the peer-review filter due to congestion.

and a saturated screening capability of $\sigma \approx 0.6$, the quality stabilizes at $Q_{corrupt} \approx \frac{\alpha}{\alpha+(1-\alpha)(1-\sigma)} \approx 0.38$. This analysis highlights a profound structural fragility: the system provides no early warning signs of impending failure. A journal can appear to be in perfect health with high-quality output while sitting immediately adjacent to the tipping point, where a minor increase in submissions or a slight reduction in budget can trigger an instantaneous and irreversible collapse in reputation.

A Supplementary Integrated Analysis

In this appendix, we present unified visualizations that integrate the individual parameter sensitivity analyses discussed in the main text. By plotting related variables on dual axes or within normalized ranges, we illustrate the comparative magnitude and directional impact of competing forces within the academic ecosystem.

A.1 Supply vs. Demand: Population and Capacity

Figure 6 contrasts the opposing effects of the demand-side pressure (Total Population N) and the supply-side constraint (Journal Capacity K). The dual x-axis format highlights a critical dynamic: increasing the population drives the system linearly toward corruption (Blue line), whereas increasing capacity provides a stabilizing force that restores honesty (Red dashed line). The intersection of these trends defines the sustainability limit of the journal.

A.2 Normalized Parameter Sensitivity: Productivity, Friction, and Risk

Figure 7 integrates the sensitivity analysis for three parameters defined on the unit interval $[0, 1]$: Author Productivity (α), Screening Friction (ϵ), and False Positive Rate (λ). This visualization isolates the distinct functional forms of instability: ϵ introduces linear degradation, α exhibits hyperbolic growth characteristic of the "Productivity Paradox," and λ demonstrates a non-linear phase transition from unconditional honesty to rapid corruption.

A.3 Economic Forces: Incentive vs. Deterrent

Figure 8 visualizes the fundamental tug-of-war between the economic drivers of the system. The Benefit of Acceptance (B) serves as the primary incentive for opportunistic behavior, pulling the equilibrium x^* upward into the corruption zone. Conversely, the Penalty for Rejection (D) acts as the deterrent,

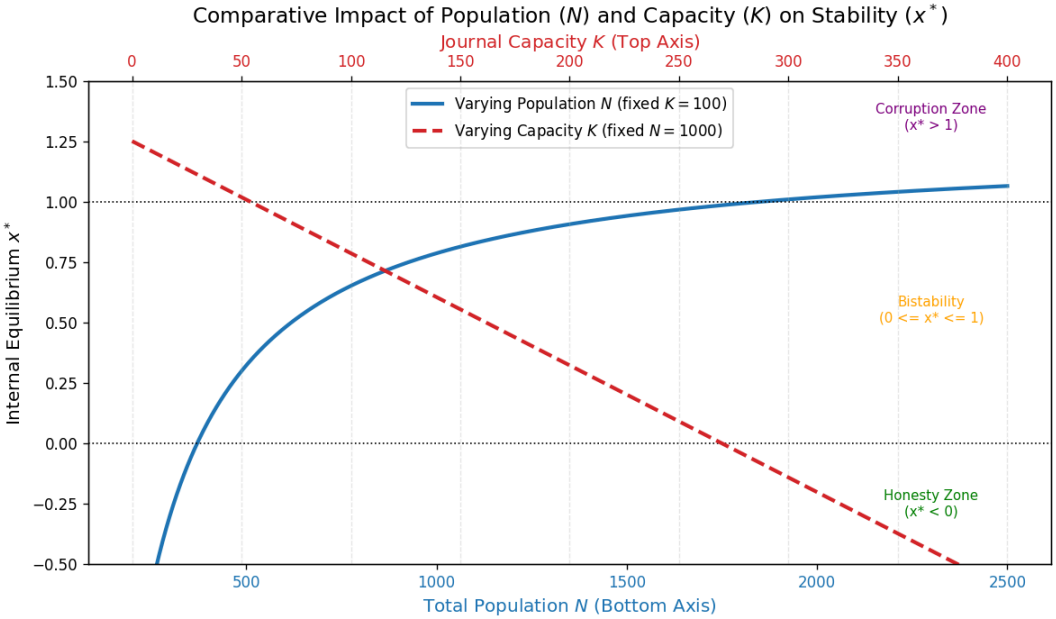


Figure 6: Comparative impact of Total Population (N, Bottom Axis) and Journal Capacity (K, Top Axis) on the internal equilibrium x^*. The opposing slopes illustrate how population growth acts as a destabilizing force (x^* increases), while capacity expansion acts as a stabilizing force (x^* decreases).

suppressing x^* back toward the honesty zone. The graph demonstrates that maintaining stability requires the penalty D to scale proportionally with the benefit B.

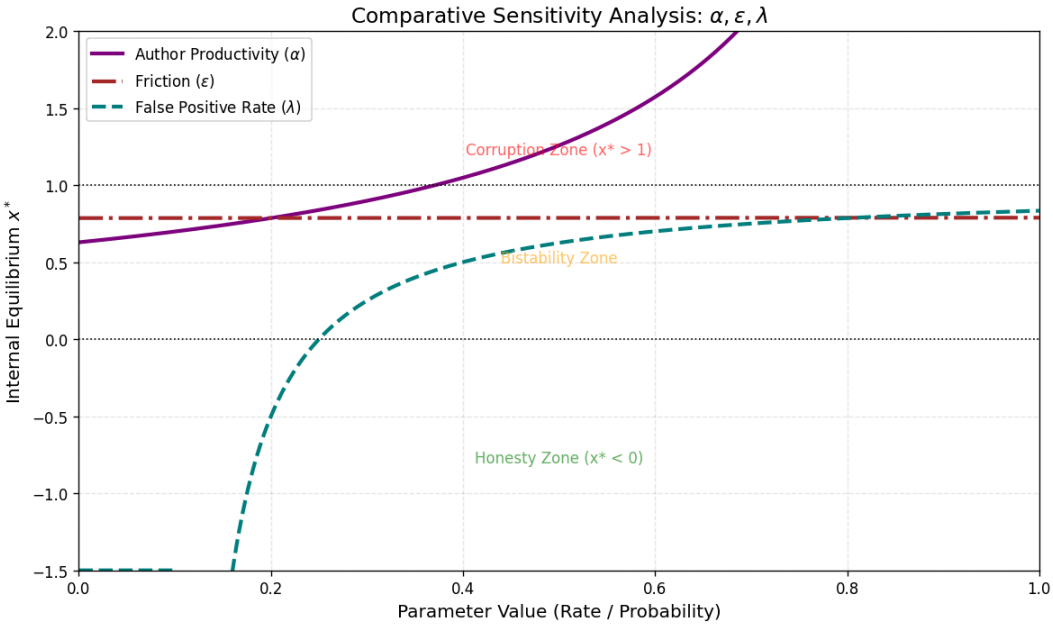


Figure 7: Unified sensitivity analysis for normalized parameters. The plot compares the linear impact of friction (ϵ, Brown), the hyperbolic impact of author productivity (α, Purple), and the threshold-based phase transition triggered by the false positive rate (λ, Teal).

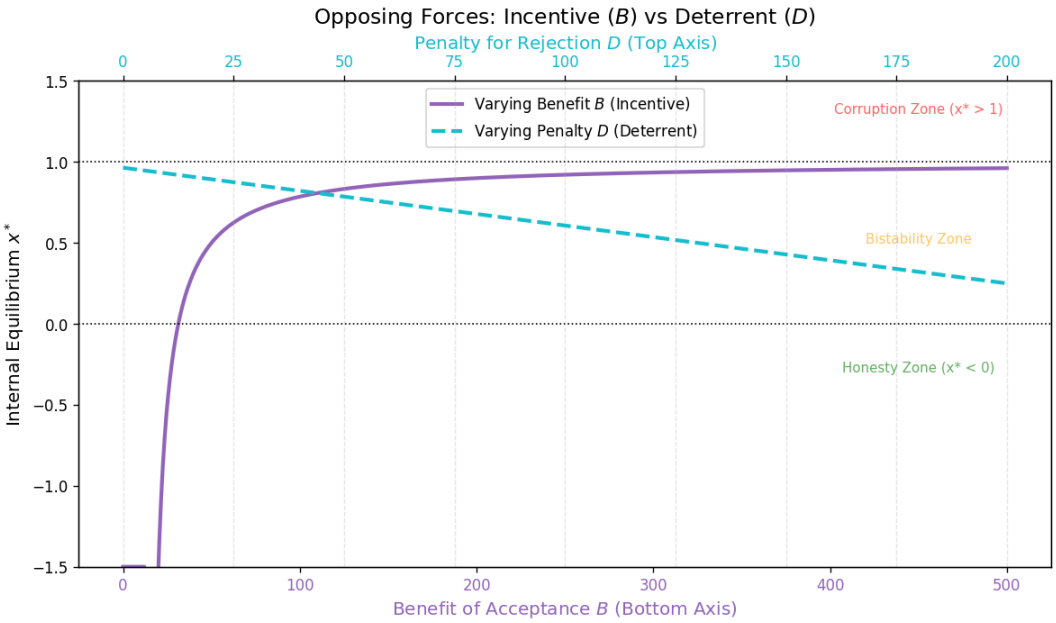


Figure 8: Comparative analysis of Economic Forces using a dual x-axis. The Purple line (Bottom Axis) shows how increasing the Benefit (B) destabilizes the system, while the Cyan dashed line (Top Axis) illustrates the stabilizing effect of increasing the Penalty (D).

A Model Parameters

The following table summarizes the key parameters used in the evolutionary game model, consistent with the empirical calibration and theoretical definitions discussed in the main text.

Symbol	Description	Role in Model
N	Total number of authors	Determines congestion intensity
K	Journal capacity	Sets the physical limit on publications
α	Good paper probability	Intrinsic productivity of authors
ϵ	False negative rate	Probability good papers are rejected
λ	False positive rate	Probability bad papers are accepted
r	Author reward	Incentive for publication
c	Submission cost	Barrier to entry for authors
B	Journal benefit	Utility from publishing good science
D	Journal penalty	Reputational cost of bad science
ρ	Rationing probability	Endogenous probability of acceptance

Table 3: Key notation and definitions for the Author-Journal Game.

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